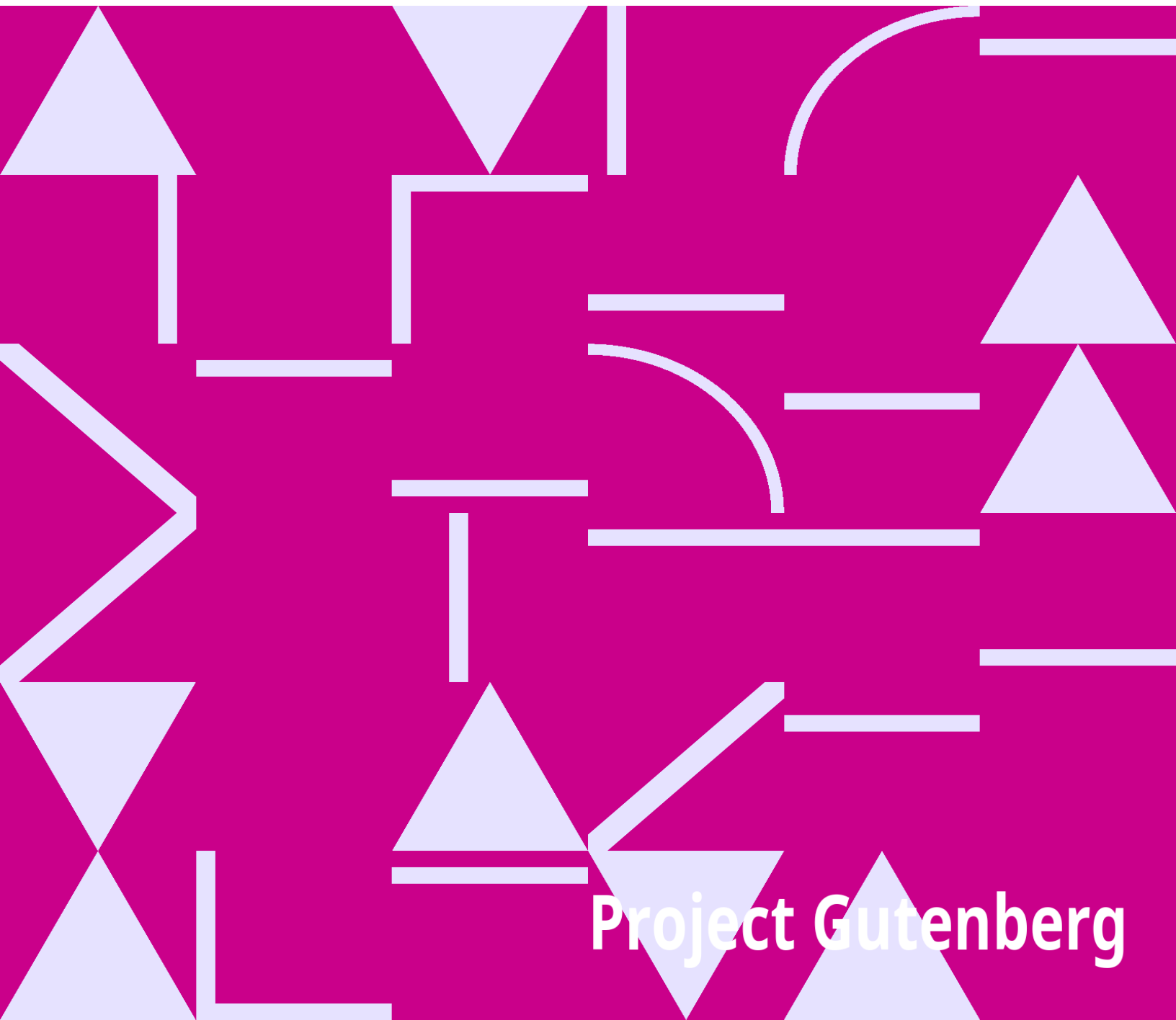


A Study of American Beers and Ales

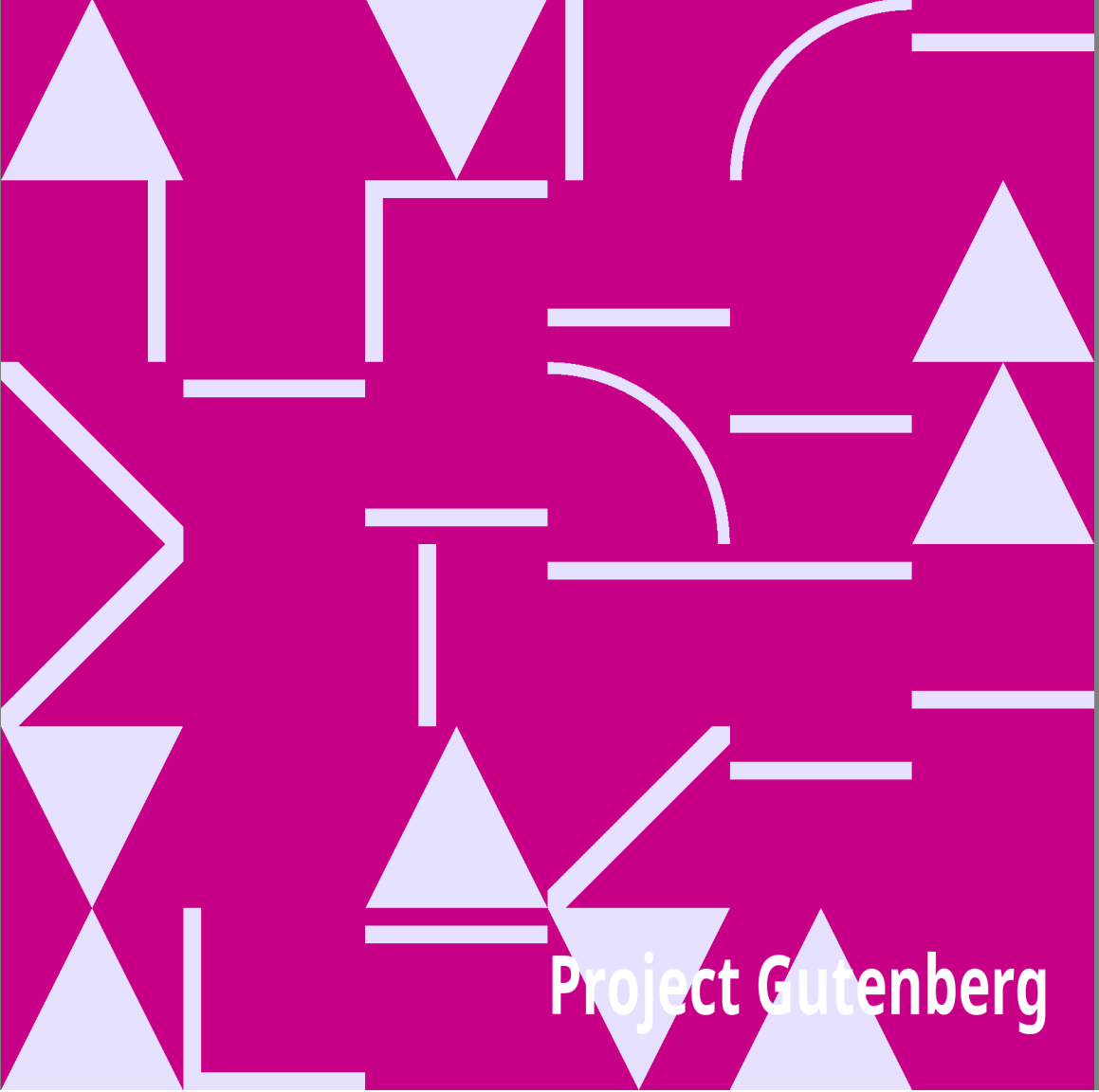
James Garfield Riley and L. M. Tolman



Project Gutenberg

A Study of American Beers and Ales

James Garfield Riley and L. M. Tolman



Project Gutenberg

The Project Gutenberg eBook of A Study of American Beers and Ales

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: A Study of American Beers and Ales

Author: James Garfield Riley
L. M. Tolman

Release date: April 11, 2008 [eBook #25050]

Language: English

Other information and formats: www.gutenberg.org/ebooks/25050

Credits: Produced by Bruce Thomas, Sigal Alon and the Online
Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK A STUDY OF
AMERICAN BEERS AND ALES ***

Transcriber's Note	This title has several large tables that may not always display well. For reference purposes, there are links to cleaned up scans for these tables.
-----------------------	---

UNITED STATES DEPARTMENT OF AGRICULTURE



BULLETIN No. 493



Contribution from the Bureau of Chemistry
CARL L. ALSBERG, Chief

Washington, D. C. PROFESSIONAL PAPER March 21, 1917

A STUDY OF AMERICAN BEERS AND ALES.

By L. M. TOLMAN, *Chief, Central Inspection District*, and J. GARFIELD
RILEY, *Assistant Chemist, Food and Drug Inspection Laboratory, New York,*
N. Y.

CONTENTS.

	Page.
Introduction	<u>1</u>
Method of undertaking the investigation	<u>2</u>
Methods of analysis	<u>3</u>
Results of analysis	<u>3</u>
Effect of raw materials used upon composition of the finished brew	<u>11</u>
Conclusions	<u>23</u>

INTRODUCTION.

The investigation, the results of which are reported in this bulletin, was undertaken for the purpose of securing information in regard to the composition of brewery products made in this country. The main object of this investigation was to find, if possible, a means of distinguishing beers and ales made entirely from malt from those made from malt together with other cereal products, such as rice, corn, and cerealin.

It was concluded, after looking into the literature, that in order to accomplish this purpose it would be necessary to collect a series of samples made from the various raw materials ordinarily used and make a study of the effect of these raw materials upon the composition of the finished product.

The investigation seemed desirable for the reason that practically all of the existing data related to foreign beers, in the preparation of which a type of malt was used entirely different from that ordinarily used in the production of American beers. Furthermore, very few of the existing data relating either to foreign or domestic beers were based upon samples concerning which exact information was available in regard to the raw materials used in the wort.

METHOD OF UNDERTAKING THE INVESTIGATION.

It was felt that it would be wholly unsatisfactory to make this investigation by means of laboratory brewings on a small scale, as the results thus obtained would not show the true conditions, because it is not possible in the laboratory to duplicate exactly the mashing or fermenting processes actually used in a commercial way. It was decided, therefore, to attempt, with the cooperation of several breweries, to make this study under the exact conditions prevailing in commercial plants. Access was secured to several breweries making different types of products from various kinds of raw materials, under such conditions that it was possible to obtain a complete history of the beer through its various stages to the finished product. One of the writers (Riley) watched the method of manufacture during its whole process and obtained samples of the product at the various stages of manufacture. Thus, it was possible to procure finished samples with practically the same degree of certainty, as regards knowledge of composition and history, as would have been the case had they been prepared in the laboratory.

In three different breweries manufacturing a wide range of products samples of the wort and beer were obtained in this manner, the entire process of manufacture being studied in detail. A record showing the kind and amount of raw materials placed in the mash and in the cooker was made of the samples collected from these three breweries. A record also was kept of the time and temperature of each operation until the mash was ready to run into the kettle. The filtering and sparging ^[1] of the mash, the time of boiling in the kettle, the amount of hops added and the point at which they were added, and the break ^[2] of the wort were all noted. After the wort had been pumped from the kettle its course was followed through the hop jack ^[3] over the coolers to the settling tank. The specific gravity or

Balling ^[4] of the original wort, the temperature at which the product was pitched, ^[5] the aeration of the wort, the kind and amount of yeast added, as well as the time and maximum temperature of the primary fermentation, also were noted. The course of the beer through the storage vats, chip casks, and filters to the racks was watched, and samples of the wort and of the beer in its various stages of production were collected and examined.

^[1] Washing the grains with hot water to remove the extract or valuable constituents as completely as possible.

^[2] Precipitation and uniting, in the form of flakes, of the coagulable albuminoids, leaving the liquid clear.

^[3] A filtering tank.

^[4] Percentage of solids in the liquor according to the Balling hydrometer.

^[5] Pitching is the operation of adding the yeast to the wort.

METHODS OF ANALYSIS.

The methods of analysis used were those given in Bulletin 107, revised (U. S. Dept. Agr., Bur. Chem.), pages 90-94, with the exception that the determination of phosphoric acid was made by the method used in fertilizer analysis (*ibid.*, pp. 2-5), destroying the organic material in the beer by digestion with strong sulphuric acid and nitric acid and determining the phosphoric acid finally by the optional volumetric method (*ibid.*, p. 4). The uranium acetate method given for beers was not used, for the reason that it was found to be exceedingly difficult to obtain accurate results on dark-colored beers. ^[1]

^[1] Riley, in his report to the Association of Official Agricultural Chemists for the year 1913, stated that the method giving the most uniform results was that of ashing the beer with an excess of standard calcium acetate, and that while the moist combustion method in the hands of those familiar with it gave satisfactory results, the various collaborators working with the method did not get as uniform results as with the method of ashing with calcium acetate. *J. Assoc. Off. Agr. Chemists* 1 (1915), 138-143.

It was found in the estimation of dextrin by the Sachsse-Allihn method (*ibid.*, p. 91) that there is an error in the method of calculation of the amount of dextrose formed from the amount of maltose in the original beer. Instead of multiplying the amount of maltose in the original beer by the factor 0.9, it should be multiplied by the factor 1.053, as 1 gram of anhydrous maltose yields, on hydrolysis, 1.053 grams of dextrose. The product is the quantity which should be subtracted from the total amount of dextrose found after hydrolysis. The extract in the beer was determined by use of the tables of Schultz and Ostermann (*ibid.*, pp. 209-213). The same methods were used in the analyses of the worts as were used in the examination of the beers.

RESULTS OF ANALYSIS.

Tables I to IV contain the results of the analyses of the worts and finished fermented products obtained at the various breweries where this investigation was conducted, arranged so as to show readily the changes which took place during fermentation and, in a few cases, the changes which took place during storage. The results are all given in terms of grams per 100 cc, so that a direct comparison of the quantities of any particular ingredient in a definite volume of material may be made. The comparison of the grams per 100 cc of an ingredient in the wort, with the grams per 100 cc in the finished fermented product, is based on the assumption that there is no appreciable change in the volume of the wort during fermentation.

In Table I are given the results of the analyses of 7 malt worts and the beers produced from them. Table II contains the results of the analyses of 2 malt-and-rice worts and 2 malt-and-corn worts, and the beers produced from them. In Table III are given the results of the analyses of 4 porter worts and the finished porters produced from them. The results of the analyses of 9 ale worts and the finished ales are shown in Table IV. In these four tables the extract in the original wort has been calculated by multiplying the alcohol (expressed in terms of grams per 100 cc) by 2, and adding to the product the extract of the beer, porter, or ale (expressed in terms of grams per 100 cc). In the porter and ale worts a percentage of dextrose had been added as brewer's sugar. Since dextrose reduces more copper than does maltose in the determination of the sugars, in order to obtain the true percentage of total sugars it was necessary to calculate the amount of copper reduced by the known amount of dextrose present, and then to calculate the amount of maltose. The results thus obtained are given in Tables III and IV under the heading "Reducing sugars as anhydrous maltose."

TABLE I.—*Analyses of all-malt worts and of the beers made from them.*

Sample No.	Product.	Date of Taking Sample.	Specific Gravity at 15.6°C./15.6°C.	Alcohol.	Extract.	Extract in original wort (calculated).	Degree of fermentation.	Total acids as lactic.	Volatil acids as acetic
		1911.		<i>Grams per 100 cc.</i>	<i>Grams per 100 cc.</i>	<i>Grams per 100 cc.</i>		<i>Grams per 100 cc.</i>	<i>Gram per 100 cc.</i>
22013-D	Wort	July 5	1.0518	...	13.75	...	...	0.198	0.00
22017-D	Beer	July 12	1.0125	3.85	5.16	12.86	59.88	.216	.00
22014-D	Wort	July 6	1.0517	...	13.71	...	...	.198	.00
22018-D	Beer	July 13	1.0124	3.91	5.16	12.98	60.25	.225	.00
...	Wort	July 7	1.0517	...	13.71	...	...	.207	.00
22019-D	Beer	July 14	1.0135	3.83	5.44	13.10	58.47	.234	.00
22015-D	Wort	July 8	1.0517	...	13.70	...	...	.198	.00
22020-D	Beer	July 15	1.0130	3.72	5.22	12.66	58.77	.236	.00
22016-D	Wort	July 10	1.0515	...	13.68	...	...	.225	.00
22021-D	Beer	July 17	1.0147	3.66	5.66	12.98	56.39	.221	.00
		1912.							
16289-C	Wort	Feb. 29	1.0455	...	12.05	...	...	.216	.00
16289-C	Beer	Mar. 7	1.0171	3.02	6.01	12.11	49.88	.230	.00
20714-D	Do	Apr. 18	1.0167	3.12	5.90	12.14	51.40	.243	.01
16299-C	Wort	Mar. 1	1.0454	...	12.02	...	...	.180	.00
16299-C	Beer	Mar. 18	1.0180	2.87	6.16	11.94	48.07	.234	.01

TABLE II.—*Analyses of malt-and-rice and malt-and-corn worts and of the beers made from them.*

Sample No.	Raw Materials.	Product.	Date of Taking Sample.	Specific Gravity at	Alcohol.	Extract.	Extract in original	Deg ferme
------------	----------------	----------	------------------------	---------------------	----------	----------	---------------------	-----------

				15.6°C./ 15.6°C.			wort (calculated).	
					Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	
22026-D	80 per cent malt and 20 per cent rice.	Wort	July 24 1911	1.0459	...	12.15	...	
22032-D	Do	Beer	July 31 1911	1.0121	3.31	4.80	11.42	
22036-D	Do	Wort	Aug. 9 1911	1.0464	...	12.30	...	
22042-D	Do	Beer	Aug. 16 1911	1.0139	3.18	5.23	11.59	
16269-C	60 per cent malt and 40 per cent corn.	Wort	Dec. 5 1911	1.0489	...	12.95	...	
16271-C	Do	Beer	Dec. 12 1911	1.0149	3.45	5.68	12.58	
16287-C	Do	do	Mar. 12 1912	1.0159	3.33	5.76	12.42	
16270-C	Do	Wort	Dec. 6 1911	1.0496	...	13.14	...	
16272-C	Do	Beer	Dec. 13 1912	1.0152	3.41	5.61	12.43	
16286-C	Do	do	Mar. 12 1912	1.0157	3.29	5.68	12.26	

TABLE III.—Analyses of porter worts made from malt, cerealin, and brewer's sugar, and of the porters made from these worts.

Sample No.	Product.	Date of Taking Sample.	Specific Gravity at 15.6°C./ 15.6°C.	Alcohol.	Extract.	Extract in original wort (calculated).	Degree of fermentation.	Total acids as lactic.	Volatile acids as acetic.	Re su anl m
		1911.			Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	C 1
22023-D	Wort	July 20	1.0572	...	15.25	...	...	0.270	0.002	
22028-D	Porter	July 26	1.0151	4.30	6.02	14.62	58.82	.324	.003	
22046-D	do	Aug. 21	1.0135	4.48	5.70	14.66	61.12	.380	.002	
22038-D	Wort	Aug. 11	1.0625	...	16.66	...	...	.234	.002	
22044-D	Porter	Aug. 18	1.0165	4.75	6.60	16.10	59.01	.342	.002	
22045-D	Wort	Aug. 18	1.0623	...	16.62	...	...	.234	.002	
29501-B	Porter	Aug. 25	1.0170	4.72	6.72	16.16	58.42	.288	.002	
22025-D	Wort	July 21	1.0633	...	16.87	...	...	.234	.004	
22031-D	Porter	July 28	1.0178	4.70	6.96	16.36	57.46	.378	.002	

22049-D	do	Aug. 24	1.0162	4.82	6.54	16.18	59.58	.306	.003	
---------	----	---------	--------	------	------	-------	-------	------	------	--

TABLE IV.—*Analyses of ale worts and of the ales made from them.*

Sample No.	Product.	Raw Materials.	Date of Taking Sample.	Specific Gravity at 15.6°C./15.6°C.	Alcohol.	Extract.	Extract in original wort (calculated).	De-ferment
					<i>Grams per 100 cc.</i>	<i>Grams per 100 cc.</i>	<i>Grams per 100 cc.</i>	
22024-D	Malt, cereal, and brewer's sugar	Wort	1911. July 21	1.0608	...	16.24	...	
22030-D	do	Ale	July 27	1.0139	4.82	5.93	15.57	
22047-D	do	Ale after storage.	Aug. 22	1.0124	4.97	5.62	15.56	
22034-D	do	Wort	Aug. 7	1.0610	...	16.30	...	
22039-D	do	Ale	Aug. 14	1.0123	5.11	5.67	15.89	
29504-B	do	Ale after storage	Oct. 6	1.0106	5.26	5.28	15.80	
22037-D	do	Wort	Aug. 10	1.0611	...	16.32	...	
22043-D	do	Ale	Aug. 17	1.0124	4.93	5.62	15.48	
22022-D	Malt and cereal	Wort	July 19	1.0642	...	17.10	...	
22027-D	do	Ale	July 26	1.0133	5.17	5.95	16.29	
22035-D	do	Wort	Aug. 8	1.0668	...	17.80	...	
22040-D	do	Ale	Aug. 15	1.0145	5.34	6.30	16.98	
16267-C	do	Ale after storage	Nov. 15	1.0134	5.52	6.11	17.15	
29506-B	do	Wort	Oct. 10	1.0760	...	20.23	...	
29512-B	do	Ale	Oct. 17	1.0253	5.43	9.19	20.05	
13922-D	do	Ale after storage	1912. Jan. 3	1.0208	5.80	8.18	19.78	
29507-B	do	Wort	1911. Oct. 11	1.0767	...	20.40	...	
29514-B	do	Ale	Oct. 18	1.0228	5.67	8.52	19.86	
29513-B	do	Wort	Oct. 18	1.0781	...	20.75	...	
29519-B	do	Ale	Oct. 24	1.0244	5.53	9.00	20.06	
13923-D	do	Ale after storage	1912. Jan. 2	1.0210	5.67	8.18	19.52	
22050-D	do	Wort	1911. Aug. 23	1.0793	...	21.05	...	

A study of these tables shows very clearly that during fermentation marked changes are brought about other than the mere conversion of sugar into alcohol. While it is well known that these changes take place it seems worth while to consider them here, because no similar study relating to American brewery products has been published. Further since we have the exact analysis of the wort and of the beer which was made from it, we have a special opportunity to examine quantitatively some of these changes, such as the production of alcohol, the fermentation of dextrin, the development of acids, and the losses of protein, ash, and phosphoric acid during fermentation.

In order to study the question of the yield of alcohol, to test the present factor used for the calculation of the solids in the original wort, and to show the approximate amount of dextrin, calculations were made, the results of which are presented in Table V.

TABLE V.—*Changes taking place in the conversion of worts into beers and ales.*

Product.	Loss in solids.	Loss in sugar.	Alcohol.	Loss in solids divided by alcohol.	Difference between loss in solids and loss in sugar.
	<i>Grams per 100 cc.</i>	<i>Grams per 100 cc.</i>	<i>Grams per 100 cc.</i>		<i>Grams per 100 cc.</i>
Beer (all-malt)	8.59	8.45	3.85	2.23	0.14
Do	8.55	8.43	3.91	2.18	.12
Do	8.27	8.49	3.83	2.13	.22
Do	8.48	8.27	3.72	2.27	.21
Do	8.02	7.99	3.66	2.19	.03
Beer (60 per cent malt and 40 per cent corn)	7.27	7.42	3.45	2.10	.15

	Do	7.53	7.73	3.33	2.00	.20
Beer (80 per cent malt and 20 per cent rice)		7.35	7.46	3.31	2.22	.11
	Do	7.07	7.13	3.18	2.22	.06
Beer (all-malt)		6.04	6.05	3.02	2.00	.01
	Do	5.86	6.01	2.87	2.04	.15
Average for beers					2.14	.04
Porter (small)		9.23	7.60	4.30	2.14	1.63
Porter (large)		9.91	8.32	4.70	2.10	1.59
	Do	10.06	8.27	4.75	2.11	1.79
	Do	9.90	7.82	4.72	2.09	2.08
Ale		10.31	8.84	4.82	2.13	1.47
	Do	10.63	8.98	5.11	2.08	1.65
	Do	10.70	9.20	4.93	2.17	1.50
	Do	11.15	8.94	5.16	2.15	2.21
	Do	11.50	9.57	5.34	2.15	1.93
	Do	11.62	9.06	5.35	2.17	2.56
Pale ale		11.04	9.50	5.43	2.03	1.54
	Do	11.88	9.63	5.67	2.09	2.25
	Do	11.75	9.11	5.53	2.12	2.64
Brown stout		12.03	9.38	5.69	2.11	2.65
Average for ales		...	...	...	2.12	1.96
Average for beers and ales		...	...	...	2.13	...

In Table V have been collected results (calculated from Tables I-IV) which show the loss in solids between the wort and the finished fermented product, the loss in sugar, the yield of alcohol, the loss in solids divided by the alcohol, and the difference between the loss in solids and the loss in sugar. By dividing alcohol into loss in solids there was secured a factor which makes possible the estimation of the solids in the original wort, provided that alcohol and extract are known. This factor also shows the yield of alcohol for a given amount of solids disappearing during fermentation. It has been found in the case of the beers that this factor averages 2.14, while in the case of the ales it averages 2.12, making an average for all of the products of 2.13. This clearly shows that in the yield of alcohol for a given amount of fermentable solids there is no appreciable difference between top fermentation products, such as ales, and bottom fermentation products, such as beers.

A marked difference in loss in solids is shown, however, when we compare the beers with the ales. In the case of the beers we find there is practically no difference between the loss in solids and the loss in sugar, while in the case of the porters and ales there is a very appreciable difference. The difference between the loss in solids and the loss in sugar is only 0.04 per cent for all of the beers; while in the case of the porters and ales the difference varies from 1.47 per cent to 2.65 per cent, with an average of 1.96 per cent. These figures clearly show that in the case of the porters and ales there has been some material other than sugar fermented. Unfortunately, the determination of dextrin was not made in all of the worts, so that the actual decrease in dextrin can be shown only in a few cases. But in those cases where we have the actual results the difference between loss in solids and the loss in sugar compares very closely with the actual amount of dextrin disappearing during fermentation.

DEVELOPMENT OF ACIDS DURING FERMENTATION.

A comparison of the amounts of volatile and fixed acids in the worts and in the finished beers shows that normally there is no appreciable development of volatile acid during fermentation and only a slight increase in the fixed acid. This increase in fixed acid averages in the case of the beers 0.049 per cent, while in the case of the ales the increase averages 0.103 per cent.

DECREASE IN PROTEIN, ASH, AND PHOSPHORIC ACID.

A general study of the preceding tables will show that there is an appreciable loss of protein, ash, and phosphoric acid during the fermentation. Table VI has been prepared to show the average loss during fermentation of the various classes of worts with respect to their protein, ash, and phosphoric acid contents.

TABLE VI.—Average loss during fermentation.

Kind of wort.	Protein.	Ash.	Phosphoric acid.
	<i>Grams per 100 cc.</i>	<i>Grams per 100 cc.</i>	<i>Grams per 100 cc.</i>
Beer worts (all-malt)	0.209	0.017	0.015
Beer worts (malt and rice)	.210	.045	.014
Beer worts (malt and corn)	.168	.014	.013
Porter worts	.213	.031	.017
Ale worts	.275	.029	.019

The results given in Table VI show a great similarity in the changes in all of the products, as there is about the same amount of loss of protein, ash, and phosphoric acid in the beer, ale, and porter worts. There does not appear to be any appreciable loss, however, of either protein or phosphoric acid during the storage or aging period as is shown by the few samples which we have analyzed after storage. This is practically in agreement with the experiment of Bertschinger, [\[1\]](#) whose results show only a very slight increase in alcohol and loss of sugar during the storage period.

[\[1\]](#) Z. angew. Chem. (1890), p. 670.

EFFECT OF RAW MATERIALS USED UPON COMPOSITION OF THE FINISHED BREW.

In order to show the effects on the finished beers or ales of the use of corn, rice, cerealine, and brewer's sugar as substitutes for malt in the worts, Table VII has been prepared, giving the results of analyses of a number of brews made in different breweries and from varying kinds and amounts of raw materials.

TABLE VII.—*Analyses of beers and ales from various breweries.*

Sample No.	Raw materials.	Product.	Alcohol.	Extract (Schultz and Ostermann).	Extract in original wort (calculated).
	<i>Brewery No. 1.</i>		<i>P.ct. by wght.</i>	<i>P.ct.</i>	<i>P.ct.</i>
14004-H	Malt	Beer	3.07	5.55	11.69
14005-H	65 per cent malt and 35 per cent cerealine	do	2.42	6.19	11.03
14006-H	60 per cent malt and 40 per cent corn	do	2.16	5.28	9.60
	<i>Brewery No. 2.</i>				
22017-D	Malt	Beer	3.83	5.06	12.72
22018-D	do	do	3.90	5.06	12.86
22020-D	do	do	3.69	5.12	12.50
22021-D	do	do	3.63	5.54	12.80
22042-D	80 per cent malt and 20 per cent rice	do	3.16	5.13	11.45
	<i>Brewery No. 3.</i>				
29517-B	80 per cent malt and 20 per cent cerealine	Ale	6.33	6.77	19.43
22027-D	78 per cent malt and 22 per cent cerealine	do	5.14	5.82	16.10

22040-D	do	do	5.31	6.17	16.79
22048-D	do	do	5.33	6.15	16.81
29512-B	75 per cent malt and 25 per cent cerealine	do	5.32	8.88	19.52
29514-B	do	do	5.43	8.25	19.11
29519-B	do	do	5.43	8.70	19.56
22030-D	65 per cent malt, 28 per cent cerealine, and 7 per cent brewer's sugar	do	4.79	5.80	15.38
22039-D	do	do	5.10	5.55	15.75
22043-D	do	do	4.91	5.50	15.32

[1] Brewer's scale.

In the results given under brewery No. 1, a beer made entirely from malt is compared with a beer made from 65 per cent of malt and 35 per cent of cerealine, and with a beer made from 60 per cent of malt and 40 per cent of corn, in all of which the same quality of malt was used.

In the case of brewery No. 2, a beer made entirely from malt and a beer made from 80 per cent of malt and 20 per cent of rice are given, in both of which the same quality of malt was used.

Under brewery No. 3 are given determinations for ales prepared from 80 per cent of malt and 20 per cent of cerealine; 78 per cent of malt and 22 per cent of cerealine; 75 per cent of malt and 25 per cent of cerealine; and 65 per cent of malt, 28 per cent of cerealine, and 7 per cent of brewer's sugar. The same quality of malt was used in all of these brews, but the brews were of different strengths.

Table VII is given practically in two parts, the first part showing the actual results obtained by the analysis of the finished beer or ale and the second part showing protein, ash, and phosphoric acid calculated to the basis of a uniform wort containing 15 per cent of solids.

Taking into consideration the actual results obtained upon the beers and ales, it will be seen in the case of brewery No. 1 that the three beers vary in composition to a considerable degree. Especially is this variation marked in regard to the protein, ash, and phosphoric acid contents, which exhibit a marked decrease approximately in direct proportion to the amount of cerealine or corn substituted for malt. The same condition is apparent in the case of the products made in brewery No. 2, the beer made from 80 per cent of malt and 20 per cent of rice showing a material reduction in protein, ash, and phosphoric acid. In brewery No. 3, however, a somewhat different condition is noted. Unfortunately, there is no all-malt product of this brewery to compare with the brews made from a portion of cerealine or from cerealine and brewer's sugar. It will be noted, however, that when the actual results obtained on the finished products of this brewery are compared with those of the all-malt brews of breweries Nos. 1 and 2, they do not clearly show a reduction of protein and ash as might be expected. For example, in

the case of one of the samples of the ale made with 25 per cent of cereal in and 75 per cent of malt (sample No. 29512-B), the percentage of protein is 0.65 and of the ash 0.266. The percentages of protein and ash for the three samples of this ale represented by Nos. 29512-B, 29514-B, and 29519-B are higher than were found in any of the all-malt products of the first two breweries under consideration. This, however, can be readily explained when it is considered that in the case of brewery No. 2 in the all-malt beers (sample No. 22017-D) only 58 pounds of malt were used in the preparation of a barrel of beer containing 31 gallons; while in the case of sample No. 29512-B there were used, in preparing a barrel of similar capacity, 68 pounds of malt and 23 pounds of cereal in. That is, in the second product there is, in the same volume of liquid, the extractive material from 68 pounds of malt and 23 pounds of cereal in, while in the first product there is present the extractive material from only 58 pounds of malt. Since the analysis is made upon the finished liquid it is evident that the percentage composition of any particular ingredient should be very much larger in the second product because of the very much larger amount of material used in its preparation. It is apparent, therefore, that no direct comparison can be made between the percentage composition of these different brews in order to determine the effects of the raw materials upon their composition.

The most satisfactory way to have tested this question of the effect of raw materials on the finished product would have been to make a series of worts with exactly the same percentage of solids, some of pure malt and others of mixtures of pure malt and corn, rice, and cereal in; then a direct comparison between the results would have shown the effects of these various materials. This method was impracticable because it was necessary to take the brews as actually made under varying commercial conditions. The object sought can be accomplished, however, by calculating the results of these analyses either to the basis of dry material in the original wort or by calculating them to the basis of a wort with constant water content. It was decided to calculate all of the results to the basis of a wort containing 15 per cent of solids, as this would give a uniform basis for comparison and would be approximately an average wort. The method employed in calculating the various beers and ales to this uniform basis was as follows:

The percentage of solids in the original wort was calculated by multiplying the percentage by weight of alcohol by 2 and adding the percentage by

weight of extract. The result for an ordinary beer would be about 12 per cent, while in the case of a very heavy ale it might be as high as 18 or 20 per cent. The actual percentages of protein, ash, and phosphoric acid found by analysis were then calculated to the basis of a uniform wort containing 15 per cent of solids. This was the method used for preparing the second part of this table. A study of this portion of the table shows the actual effects of the various substitutes used for malt on the composition of the fermented product. For instance, the first of the all-malt beers from brewery No. 2 (22017-D) showed in the analysis of the original product a protein percentage of 0.603, an ash percentage of 0.206, and a phosphoric acid percentage of 0.079. When calculated to the basis of a wort containing 15 per cent of solids instead of 12.72 per cent (the actual percentage of solids in the wort from which it was made), it gave the following percentages: Protein, 0.712; ash, 0.243; and phosphoric acid, 0.093. In the case of brewery No. 3, sample No. 29512-B, where the original analysis of the product showed 0.650 per cent of protein, 0.266 of ash, and 0.057 of phosphoric acid, it will be found that when this product is calculated to the basis of a wort of 15 per cent of solids instead of a wort of 19.52 (the actual percentage of solids in the wort in this case) the percentage of ash is 0.204, of protein 0.499, and of phosphoric acid 0.044. A comparison of these results shows that in the protein, ash, and phosphoric acid there has been a material reduction below the figures found upon the all-malt beer, due to the presence of the 25 per cent of cerealine. A study of these results, calculated to the basis of 15 per cent of solids in the wort, shows very clearly that the general effect of the substitution of cerealine, brewer's sugar, rice, and corn is to reduce the content of ash, protein, and phosphoric acid.

It is evident from the results here given that the most important things to be considered in judging the nature of the raw materials used in the preparation of a beer are the quantities of protein, phosphoric acid, and ash; as the other constituents present in the finished beer are more or less variable, the quantities present depending upon the methods of mashing and fermentation.

Table VIII contains a summary of results giving the ash, protein, and phosphoric acid in all of the finished products of known composition which were examined, calculated to the basis of a uniform wort of 15 per cent of solids.

TABLE VIII.—*Summary of the results of analyses (showing ash, protein, and phosphoric acid determinations) in all finished products of known composition, calculated to the basis of a uniform wort containing 15 per cent of solids.*

Raw materials.	Products.	Ash.	Protein (N × 6.25).	Phosphoric Acid (as P ₂ O ₅).
Malt	Beers: 21 samples	<i>Per cent.</i>	<i>Per cent.</i>	<i>Per cent.</i>
	Maximum...	0.336	1.079	0.143
	Minimum...	.230	.701	.087
	Average...	.275	.870	.109
80 per cent malt and 20 per cent rice	Beer	.202	.517	.073
66 per cent malt and 34 per cent rice	do	.198	.555	.084
62 per cent malt and 38 per cent rice	do	.205	.488	.061
55 per cent malt and 45 per cent rice	do	.148	.380	.077
50 per cent malt and 50 per cent rice	do	.167	.351	.056
	Maximum...	.205	.555	.084
70 per cent malt and 30 per cent corn	Beer	.199	.343	.057
Do	do	.188	.367	.065
68 per cent malt and 32 per cent corn	do	.150	.461	.057
Do	do	.181	.466	.062
Do	do	.164	.459	.056
60 per cent malt and 40 per cent corn	do	.215	.563	.074
Do	do	.188	.593	.076
Do	do	.223	.597	.074
45 per cent malt and 55 per cent corn	do	.145	.347	.057
	Maximum...	.223	.597	.076
65 per cent malt and 35 per cent cerealine	Beer	.192	.483	.057
80 per cent malt and 20 per cent cerealine	Ale	.215	.480	.051

78 per cent malt and 22 per cent cereal	do	.176	.455	.050
Do	do	.169	.476	.045
Do	do	.181	.502	.040
75 per cent malt and 25 per cent cereal	do	.204	.499	.044
Do	do	.196	.509	.044
Do	do	.191	.502	.043
65 per cent malt, 7 per cent brewer's sugar, and 28 per cent cereal	do	.185	.409	.037
Do	do	.175	.443	.040
Do	do	.166	.427	.041
	Maximum...	.213	.509	.051

A study of the results given in Table VIII shows that in the case of American beers the all-malt beers are higher in ash, protein, and phosphoric acid than are any of the beers made from a mixed mash of malt and other cereals. The difference is sufficiently marked to make it possible to draw a rather sharp line between the all-malt beers and the beers made from the present commercial mixtures. Take, for instance, the beers made from mixtures of malt and rice in which the proportion of rice varies from 20 to 50 per cent. It will be seen that in none of these samples is the ash, phosphoric acid, or protein so high as the minimum found in the all-malt beers. The same will be seen in the case of the malt-and-corn beers. In none of the malt-and-corn beers is the ash, protein, or phosphoric acid so high as the minimum found in the all-malt beers, and the same is true of the mixtures of malt and cereal and of malt, brewer's sugar, and cereal. This shows clearly that the commercial beers made in this country from malt and malt substitutes can be distinguished readily from all-malt beers.

When the average composition of the 21 all-malt beers examined is taken into consideration it will be seen that there is a very sharp line of demarcation between the all-malt and the malt, rice, and corn products. From the figures which were obtained upon American beers it would seem that protein as a rule is more sharply reduced by the addition of malt substitutes than is the ash or the phosphoric acid, although where corn or cereal is used there is a very marked reduction in the amount of

phosphoric acid. It would appear, therefore, from the results of this investigation that in the consideration of American beers it will be comparatively easy to draw a line between beers made solely from malt and those made from mixtures of malt with rice, corn, and other substitutes.

This conclusion is not entirely in agreement with the results which have been obtained by others upon foreign beers, in the preparation of which low protein barleys have been used. Joseph Race ^[1] has reported some interesting results of an investigation carried on for the same purpose as that for which this particular investigation was undertaken; that is, to distinguish between all-malt beers and those made from substitutes. His results do not show as sharp a reduction of the protein, but he found in his all-malt beers a very much lower percentage of total protein than was found in the malt beers of this country. He did observe, however, a material reduction of the phosphoric acid due to the use of substitutes. Unfortunately, he made his determination of phosphoric acid in the ash, and while he reports a marked difference between the phosphoric acid content of the malt beers and those made from substitutes, his total figures for phosphoric acid are much lower than those reported in this bulletin. For this reason the figures for total phosphoric acid given by him are not at all comparable with those determined by the moist combustion method, by the uranium acetate method, or by the method of ashing with calcium acetate. ^[2]

^[1] J. Soc. Chem. Ind., 27 (1908), 544-547.

^[2] Riley, in his report to the Association of Official Agricultural Chemists for the year 1913, showed that a large proportion of the phosphoric acid was ordinarily lost when the beer was directly ashed (J. Assoc. Off. Agr. Chemists, 1 (1915), 138-143). For this reason, in comparing the amount of phosphoric acid given in the literature on beers, it is very essential to know the method used for determining the phosphoric acid.

The same fact observed by Race, namely, that foreign beers are of low protein content, is shown very clearly in the published literature on European beers in general. König ^[3] gives the following results of analyses made by himself and H. Weigmann of two all-malt beers, calculated to the basis of a wort containing 15 per cent of solids:

Beer and percentage of wort.	Protein.	Ash.	Phosphoric acid.
	<i>Per cent.</i>	<i>Per cent.</i>	<i>Per cent.</i>
Pure Malt Beer:			
12 per cent wort	0.548	0.259	0.098
14 per cent wort	.457	.214	.076

From these results of König it will be seen that the protein content of these beers is considerably less than that of the beers examined by the writers. As the phosphoric acid and ash results, however, are practically the same as in American beers, it might be expected that the use of substitutes in place of the low-protein malt would not show so sharp a reduction of the protein as was found by the authors, although one would expect a reduction in phosphoric acid and ash similar to that found in American beers. This is confirmed by the results obtained by Race.

[3] König, F. J., *Chemie der Menschlichen Nahrungs- und Genussmittel*, 4th ed., v. 1, p. 1154. Berlin, 1903.

Robert Wahl [4] made parallel brewings of a high-protein barley and a low-protein barley, and from these obtained two beers which, when calculated to a uniform wort with 15 per cent of solids, showed a total protein in the beer made from the low-protein malt of 0.734 per cent, and in the beer made from the high-protein malt 1.041 per cent. This clearly indicates that where a beer is made from high-protein barley, as is the case with practically all of the beers made in this country, [5] the reduction in protein by the use of substitutes will be a valuable index to the true nature of the product. This, when taken in connection with the reduction of phosphoric acid brought about by the use of substitutes, gives two factors of value in judging American beers, to determine whether or not substitutes have been used; while in the case of beers made from low-protein barley there is practically only one factor, namely, the reduction of phosphoric acid.

[4] *Am. Brewers' Rev.*, 18 (1904), 339.

[5] Wahl, Robert. *In Am. Brewers' Rev.*, 29 (1915), 316-317.

After this rather extensive study had been made at the three breweries, the investigation was extended to include breweries in various sections of the country where different types of raw materials were used. A special effort was made to obtain authentic samples of practically all of the malt beers made in this country and also a large series of malt-and-rice and malt-and-corn beers. In Table IX have been tabulated the results obtained on all-malt beers. All of these results show practically the same condition noted in the other samples of malt beer; that is, a comparatively high protein and phosphoric acid content as compared with beers made in part from rice or corn. These malt beers show figures considerably higher in protein than those given in the literature for all-malt beers made from the low-protein malt of Europe.

TABLE IX.—*Analyses of all-malt American beers.*

Sample No.	Alcohol.	Extract (Schultz and Ostermann).	Extract in original wort (calculated).	Degree of fermentation.	Total acid as lactic.	Volatile acid as acetic.	Reducing sugars as anhydrous maltose.	Dextrin.	(l
	<i>Per cent. by weight</i>	<i>Per cent.</i>	<i>Per cent.</i>		<i>Per cent.</i>	<i>Per cent.</i>	<i>Per cent.</i>	<i>Per cent.</i>	
22017-D	3.83	5.06	12.72	60.22	0.214	0.001	1.32	2.30	
22018-D	3.90	5.06	12.86	60.65	.223	.002	1.34	2.03	
22020-D	3.69	5.12	12.50	59.04	.234	.002	1.57	2.20	
22021-D	3.63	5.54	12.80	56.72	.219	.001	1.82	2.15	
16289-C	3.00	5.88	11.88	50.50	.228	.003	1.33	2.68	
16299-C	2.84	6.02	11.70	48.55	.232	.012	1.38	2.77	
20714-D	3.07	5.80	11.94	51.42	.241	.012	1.45	2.64	
20715-D	2.95	5.77	11.67	50.56	.228	.009	1.43	2.67	
23571-E	3.68	4.44	11.80	62.45	.232	.010	1.06	1.67	
23585-E	3.60	5.04	12.24	58.82	.277	.005	1.36	1.81	
23528-E	3.28	6.36	12.92	50.80	.384	.016	1.62	2.74	
23533-E	3.41	5.48	12.30	55.45	.232	.008	1.48	2.44	
23537-E	3.80	7.26	14.86	51.35	.250	.012	2.51	2.95	

23588-E	3.16	6.11	12.43	50.84	.250	.008	1.93	2.41
23538-E	3.13	6.61	12.77	48.24	.250	.009	2.13	2.82
23589-E	3.35	6.21	12.91	51.90	.178	.017	1.78	2.87
23539-E	3.22	6.63	13.27	48.53	.312	.017	2.18	2.58
23540-E	3.93	6.77	14.63	53.73	.348	.007	2.64	1.87
23590-E	3.48	5.45	12.41	56.09	.375	.010	2.21	1.48
23541-E	3.12	5.06	11.30	55.22	.259	.004	1.69	1.27
14004-H	3.07	5.55	11.69	52.54	.223	.013	1.59	2.64

[1] Brewer's scale.

In Tables X and XI are given the results for beers made from malt and rice and from malt and corn. A study of these tables shows the same condition as was noted in the other tables giving malt-and-rice and malt-and-corn beers; that is, the beers have a lower protein and phosphoric acid content than those made entirely from malt.

TABLE X.—*Analyses of malt-and-rice American beers.*

Sample No.	Raw materials.	Alcohol.	Extract (Schultz and Ostermann).	Extract in original wort (calculated).	Degree of fermentation
		<i>Per ct. by weight</i>	<i>Per ct.</i>	<i>Per ct.</i>	
22042-D	80 per cent malt and 20 per cent rice	3.16	5.13	11.45	55.20
23527-E	66 per cent malt and 34 per cent rice	3.32	5.50	12.14	54.70
23581-E	62 per cent malt and 38 per cent rice	2.86	6.16	11.88	48.16
23587-E	55 per cent malt and 45 per cent rice	3.56	4.96	12.08	58.94
23586-E	50 per cent malt and 50 per cent rice	3.44	5.67	12.55	54.82

[1] Brewer's scale.

TABLE XI.— *Analyses of malt-and-corn American beers.*

Sample No.	Raw materials.	Alcohol.	Extract (Schultz and Ostermann).	Extract in original wort (calculated).	Degree of fermentation
		<i>Per ct. by weight</i>	<i>Per ct.</i>	<i>Per ct.</i>	
23534-E	70 per cent malt and 30 per cent corn	2.75	5.53	11.03	50.32
23535-E	do	3.03	4.85	10.91	55.55
23518-E	68 per cent malt and 32 per cent corn	3.37	5.96	12.70	53.07
23561-E	do	3.26	6.14	12.66	51.50
23572-E	do	3.37	6.07	12.81	52.62
23584-E	60 per cent malt and 40 per cent corn	3.09	4.90	11.08	55.78
23523-E	do	3.16	6.07	12.39	51.01
23660-E	do	3.26	6.07	12.59	51.79
16286-C	do	3.19	5.60	11.98	53.34
16287-C	do	3.23	5.67	12.13	53.26
23524-E	45 per cent malt and 55 per cent corn	3.43	5.75	12.61	54.40

[1] Brewer's scale.

In Table XII have been brought together the results of the examination of a large number of commercial beers of American production, which were represented to be made from malt and hops. This representation subsequently proved to be false, although exact information as to the amount or kind of substitute used is not available. These results are of value, however, in showing the general composition of American beers made from the ordinary commercial mixtures and clearly indicate that by taking into consideration the ash, protein, and phosphoric acid content it is practicable to distinguish commercial beers made in this country from malt and malt substitutes from beers made from malt alone.

TABLE XII.—*Analyses of American beers incorrectly represented to be all-malt.*

Sample No.	Alcohol.	Extract (Schultz and Ostermann).	Extract in original wort (calculated).	Degree of fermentation.	Total acid as lactic.	Volatile acid as acetic.	Reducing sugars as anhydrous maltose.	D
	<i>Per ct. by weight</i>	<i>Per ct.</i>	<i>Per ct.</i>		<i>Per ct.</i>	<i>Per ct.</i>	<i>Per ct.</i>	<i>l</i>
2417-E	3.67	4.66	12.00	61.33	0.154	0.013	1.59	
36612-E	3.10	7.55	13.75	45.09	.232	.017	2.33	
1146-E	4.23	7.35	15.81	53.51	.241	.029	2.21	
3014-E	3.26	6.74	13.26	49.17	.167	.014	2.74	
2538-E	2.86	6.69	12.41	46.09	.214	.017	3.22	
1734-E	3.55	6.55	13.65	52.02	.250	.028	1.87	
1154-E	3.55	5.62	12.72	55.82	.187	.014	1.88	
1006-E	3.58	4.74	11.90	60.17	.138	.017	1.42	
195-E	3.56	4.21	11.33	62.84	.127	.008	2.72	
5022-E	2.95	5.60	11.50	51.30	.133	.022	2.30	
1005-E	2.69	5.33	10.71	50.24	.102	.010	1.77	
194-E	3.18	4.00	10.36	61.39	.062	.017	1.37	
185-E	3.44	3.92	10.80	63.70	.141	.011	2.06	
5314-E	3.78	5.65	13.21	57.15	.151	.010	1.46	
4451-E	3.23	6.52	12.98	49.77	.228	.009	2.80	
5696-E	4.19	6.44	14.82	56.55	.268	.012	2.15	
3359-E	4.96	7.43	17.35	57.18	.321	.019	1.76	
3358-E	3.23	5.53	11.99	53.83	.121	.009	1.88	
481-E	3.87	5.20	12.94	59.82	.147	.014	1.32	
687-E	3.24	4.71	11.19	57.91	.080	.009	1.63	
193-E	3.66	5.75	13.07	56.01	.148	.017	1.78	
5023-E	3.16	5.30	11.62	54.39	.128	.005	1.95	
5318-E	3.57	5.20	12.34	57.86	.220	.003	1.50	
6715-E	3.75	7.98	15.48	48.45	.174	.006	2.55	
6716-E	3.66	7.73	15.05	48.64	.187	.009	2.37	
2388-E	2.95	5.35	11.25	52.45	.080	.012	1.64	
2352-E	3.20	4.69	11.09	57.71	.147	.011	1.99	
2770-E	3.66	5.99	13.31	55.00	.169	.019	1.82	
8705-E	2.91	5.62	11.44	50.96	.147	.017	1.86	

8706-E	3.48	4.93	11.89	58.54	.214	.022	1.83
8704-E	2.76	4.98	10.50	52.57	.160	.010	1.62
7839-E	3.51	5.38	12.40	56.61	.142	.012	1.57
5707-E	3.35	5.82	12.52	53.51	.160	.011	2.01
8171-F	3.50	5.99	12.99	53.85	.169	.013	1.92
397-E	3.73	6.71	14.17	52.65	.169	.007	1.56

[1] Brewer's scale.

The data reported in Tables X, XI, and XII give the results of analyses of commercial American beers obtained from various breweries in different parts of the United States as these beers are found on the market at the present time; hence, they are of general value for the purpose of showing the composition of American beers. These data also are of considerable interest when we compare them with data relating to American beers published by the department in 1887.[1] A comparison of these two sets of figures shows that beers made at the present time have a much lower percentage of alcohol and are made from a wort containing a much lower percentage of solids than beers made a generation ago. The average of 28 samples examined and reported in 1887 in the publication cited [1] showed an average alcohol content of 4.63 per cent by weight and solids in the original wort of 14.79 per cent, while the average of 72 beers representing the products now on the market showed an average of 3.52 per cent by weight of alcohol and solids in the original wort of 12.50 per cent. This is a reduction of 1.11 per cent by weight of alcohol and 2.23 per cent of solids in the original wort.

[1] U. S. Dept. Agr., Div. Chem., Bul. 13, 1887, pt. 3, p. 282.

CONCLUSIONS.

The all-malt beers made in this country contain higher percentages of protein than the all-malt beers made in Europe, owing to the use in this country of a barley high in protein.

The use of rice, corn or corn products, and brewer's sugar as substitutes for malt reduces the content of protein, ash, and phosphoric acid in the finished beer.

This difference, as regards the protein, ash, and phosphoric acid, is a sufficient basis for distinguishing the all-malt beers made in this country from those containing the commercial mixtures of rice, corn, cerealine, and brewer's sugar.

It is necessary to calculate analytical results to the basis of a common wort in order to interpret them properly.

ADDITIONAL COPIES
OF THIS PUBLICATION MAY BE PROCURED FROM
THE SUPERINTENDENT OF DOCUMENTS
GOVERNMENT PRINTING OFFICE
WASHINGTON, D. C.
AT
5 CENTS PER COPY

WASHINGTON: GOVERNMENT PRINTING OFFICE: 1917

Transcriber's notes:

Changed header in Table I, 7th column to read Reducing Sugars ... instead of Reducing Sugar ... to match identical headers in the other Tables

*** END OF THE PROJECT GUTENBERG EBOOK A STUDY OF
AMERICAN BEERS AND ALES ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.

TABLE I.—Analyses of all-malt worts and of the beers made from them.

Sample No.	Product.	Date of taking sample.	Specific gravity at 15.6° C./15.6° C.	Alcohol.	Extract.	Extract in original wort (calculated).	Degree of fermentation.	Total acids as lactic.	Volatile acids as acetic.	Reducing sugar as anhydrous maltose.	Dextrin.	Protein (N×6.25).	Ash.	Phosphoric acid (as P ₂ O ₅).	Undetermined.	Color (Lovibond) in 1-inch cell.
				Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.		Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Degrees, brewer's scale.
22013-D	Wort....	1911. July 5	1.0518		13.75			0.198	0.001	9.79		0.862	0.231	0.098		13.0
22017-D	Beer....	July 12	1.0125	3.85	5.16	12.86	59.88	.216	.001	1.34	2.33	.611	.209	.080	0.67	12.0
22014-D	Wort....	July 6	1.0517		13.71			.198	.001	9.79		.876	.228	.095		13.0
22018-D	Beer....	July 13	1.0124	3.91	5.16	12.98	60.25	.225	.002	1.36	2.06	.614	.201	.078	.93	13.0
.....	Wort....	July 7	1.0517		13.71			.207	.001	10.04		.852	.252	.099		13.0
22019-D	Beer....	July 14	1.0135	3.83	5.44	13.10	58.47	.234	.002	1.55		.610	.215	.082		9.0
22015-D	Wort....	July 8	1.0517		13.70			.198	.001	9.86		.841	.240	.097		13.0
22020-D	Beer....	July 15	1.0130	3.72	5.22	12.66	58.77	.236	.002	1.59	2.23	.639	.206	.081	.55	11.0
22016-D	Wort....	July 10	1.0515		13.68			.225	.001	9.84		.845	.245	.097		13.0
22021-D	Beer....	July 17	1.0147	3.66	5.66	12.98	56.39	.221	.001	1.85	2.18	.635	.208	.082	.78	10.0
16289-C	Wort....	Feb. 29, 1912.	1.0455		12.05			.216	.001	7.40		.917	.232	.095		4.0
16289-C	Beer....	Mar. 7	1.0171	3.02	6.01	12.11	49.88	.230	.003	1.35	2.73	.757	.246	.082	.83	3.0
20714-D	Do....	Apr. 18	1.0167	3.12	5.90	12.14	51.40	.243	.012	1.47	2.68	.732	.229	.090	.79	3.0
16299-C	Wort....	Mar. 1	1.0454		12.02			.180	.001	7.41		.874	.238	.101		2.0
16299-C	Beer....	Mar. 18	1.0180	2.87	6.16	11.94	48.07	.234	.012	1.40	2.82	.737	.240	.090	.96	2.0

[back](#)

TABLE II.—Analyses of malt-and-rice and malt-and-corn worts and of the beers made from them.

Sample No.	Raw materials.	Product.	Date of taking sample.	Specific gravity at 15.6° C./15.6° C.	Alcohol.	Ex-tract.	Extract in original wort (calculated).	Degree of fermentation.	Total acids as lactic.	Volatile acids as acetic.	Reducing sugars as anhydrous maltose.	Dex-trin.	Protein (N × 6.25).	Ash.	Phos-phoric acid (as P ₂ O ₅).	Unde-ter-mined.	Color (Lovibond) in 1-inch cell.
					Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.		Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Degrees, brewer's scale.
22026-D	80 per cent malt and 20 per cent rice.	Wort....	July 24, 1911	1.0459		12.15			0.124	0.001	8.64		0.714	0.193	0.069		2.0
22032-D	Do.....	Beer....	July 31, 1911	1.0121	3.31	4.80	11.42	57.97	.275	.003	1.18	2.47	.519	.150	.055	0.48	2.0
22036-D	Do.....	Wort....	Aug. 9, 1911	1.0464		12.30			.126	.001	8.70		.624	.204	.072		2.0
22042-D	Do.....	Beer....	Aug. 16, 1911	1.0139	3.18	5.23	11.59	54.87	.243	.003	1.57	2.41	.400	.156	.057	.69	2.0
16269-C	60 per cent malt and 40 per cent corn.	Wort....	Dec. 5, 1911	1.0489		12.95			.144	.003	9.05		.461	.229	.066		
16271-C	Do.....	Beer....	Dec. 12, 1911	1.0149	3.45	5.68	12.58	54.85	.171	.013	1.63	2.49	.308	.213	.056	1.04	
16287-C	Do.....	do.....	Mar. 12, 1912	1.0159	3.33	5.76	12.42	53.62	.180	.014	1.87	2.36	.327	.208	.056	.99	
16270-C	Do.....	Wort....	Dec. 6, 1911	1.0496		13.14			.144	.003	9.32		.498	.216	.067		
16272-C	Do.....	Beer....	Dec. 13, 1911	1.0152	3.41	5.61	12.43	54.87	.171	.013	1.59	2.58	.314	.203	.050	.93	
16286-C	Do.....	do.....	Mar. 12, 1912	1.0157	3.29	5.68	12.26	53.67	.175	.014	1.88	2.58	.314	.222	.058	.69	

[back](#)

TABLE III.—Analyses of porter worts made from malt, cereal, and brewer's sugar, and of the porters made from these worts.

Sample No.	Product.	Date of taking sample.	Specific gravity at 15.6° C./15.6° C.	Alcohol.	Extract.	Extract in original wort (calculated).	Degree of fermentation.	Total acids as lactic.	Volatile acids as acetic.	Reducing sugars as anhydrous maltose.	Dextrin.	Protein (N×6.25)	Ash.	Phosphoric acid (as P ₂ O ₅).	Undetermined.	Color (Lovibond) in 1-inch cell.
		1911.		Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.		Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Degrees, brewer's scale.
22023-D	Wort.....	July 20	1.0572		15.25			0.270	0.002	8.67		0.696	0.227	0.069		58.0
22028-D	Porter.....	July 28	1.0151	4.30	6.02	14.62	58.82	.324	.003	1.07	3.34	.482	.208	.052	0.92	59.0
22046-D	do.....	Aug. 21	1.0135	4.48	5.70	14.66	61.12	.380	.002	1.07	3.19	.482	.182	.051	.78	60.0
22038-D	Wort.....	Aug. 11	1.0625		16.66			.234	.002	9.74		.685	.232	.072		65.0
22044-D	Porter.....	Aug. 18	1.0165	4.75	6.60	16.10	59.01	.342	.002	1.42	3.48	.477	.196	.053	1.02	60.0
22045-D	Wort.....	Aug. 18	1.0623		16.62			.234	.002	9.68		.667	.224	.069		55.0
23051-B	Porter.....	Aug. 25	1.0170	4.72	6.72	16.16	58.42	.288	.002	1.41	3.62	.482	.201	.058	1.01	55.0
22025-D	Wort.....	July 21	1.0633		16.87			.234	.004	10.25		.721	.214	.073		70.0
22031-D	Porter.....	July 28	1.0178	4.70	6.96	16.36	57.46	.378	.002	1.40	3.68	.476	.194	.055	1.21	60.0
23049-D	do.....	Aug. 24	1.0162	4.82	6.54	16.18	59.58	.306	.003	1.44	3.44	.479	.189	.053	.99	60.0

[back](#)

TABLE IV.—Analyses of ale worts and of the ales made from them.

Sample No.	Raw materials.	Product.	Date of taking sample.	Specific gravity at 15.6° C./15.6° C.	Alcohol.	Ex-tract.	Ex-tract in original wort (calculated).	Degree of fermentation.	Total acids as lactic.	Volatile acids as acetic.	Reducing sugars as anhydrous maltose.	Dextrin.	Protein (N X 6.25).	Ash.	Phosphoric acid (as P ₂ O ₅).	Undetermined.	Color (Lovibond) in 4-inch cell.
22024-D	Malt, cereal, and brewer's sugar.	Wort.....	1911. July 21	1.0608	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.		Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.	Degrees, brewer's scale.
22030-D	do.....	Ale.....	July 27	1.0139	4.82	5.93	15.57	61.91	.225	.003	1.16	3.23	.425	.194	.039	0.92	5.0
22047-D	do.....	Ale after storage.	Aug. 22	1.0124	4.97	5.62	15.56	63.88	.270	.003	1.43	2.69	.441	.180	.038	.88	5.0
22034-D	do.....	Wort.....	Aug. 7	1.0610		16.30			.153	.003	10.24		.771	.214	.060		5.0
22039-D	do.....	Ale.....	Aug. 14	1.0123	5.11	5.67	15.89	64.32	.225	.003	1.26	2.85	.471	.186	.043	.90	5.0
29504-B	do.....	Ale after storage.	Oct. 6	1.0106	5.26	5.28	15.80	66.58	.225	.003	1.10	2.60	.462	.189	.041	.93	5.0
22037-D	do.....	Wort.....	Aug. 10	1.0611		16.32			.149	.002	10.60		.763	.209	.057		4.0
22043-D	do.....	Ale.....	Aug. 17	1.0124	4.93	5.62	15.48	63.69	.216	.003	1.40	2.82	.441	.172	.043	.79	4.0
22022-D	Malt and cereal.	Wort.....	July 19	1.0642		17.10			.180	.002	10.19		.776	.212	.072		3.0
22027-D	do.....	Ale.....	July 26	1.0133	5.17	5.95	16.29	63.47	.207	.004	1.25	3.28	.500	.192	.055	.73	3.0
22035-D	do.....	Wort.....	Aug. 5	1.0668		17.80			.189	.002	10.99		.776	.212	.072		3.0
22040-D	do.....	Ale.....	Aug. 15	1.0145	5.34	6.30	16.98	62.90	.293	.004	1.42	3.31	.541	.192	.051	.84	4.0
16267-C	do.....	Ale after storage.	Nov. 15	1.0134	5.52	6.11	17.15	64.37	.234	.004	1.65	2.76	.506	.213	.053	.98	3.0
29506-B	do.....	Wort.....	Oct. 10	1.0760		20.23			.198	.001	11.19	6.81	.959	.294	.077	.98	3.0
29512-B	do.....	Ale.....	Oct. 17	1.0253	5.43	9.19	20.05	54.16	.369	.008	2.69	4.39	.671	.273	.058	1.17	3.0
13922-D	do.....	Ale after storage.	1912. Jan. 3	1.0208	5.80	8.18	19.78	58.64	.360	.008	2.54	3.42	.622	.275	.057	1.32	4.0
29507-B	do.....	Wort.....	1911. Oct. 11	1.0767		20.40			.198	.001	12.14	5.59	.965	.282	.078	1.42	3.0
29514-B	do.....	Ale.....	Oct. 18	1.0228	5.67	8.52	19.86	57.10	.360	.005	2.51	4.08	.664	.256	.057	1.01	2.0
29513-B	do.....	Wort.....	Oct. 18	1.0781		20.75			.198	.002	11.73	5.95	.936	.300	.075	1.33	2.0
29519-B	do.....	Ale.....	Oct. 24	1.0244	5.53	9.00	20.06	55.13	.281	.006	2.62	4.31	.671	.255	.057	1.14	2.0
13923-D	do.....	Ale after storage.	1912. Jan. 2	1.0210	5.67	8.18	19.52	58.09	.360	.013	2.54	3.61	.603	.282	.064	1.15	4.0
22050-D	do.....	Wort.....	1911. Aug. 23	1.0793		21.05			.270	.003	11.41	6.34	.864	.326	.091	2.11	75.0
29503-B	do.....	Stout.....	Aug. 30	1.0242	5.69	9.02	20.40	55.78	.558	.003	2.03	4.54	.735	.286	.071	1.42	69.0

[back](#)

TABLE V.—Changes taking place in the conversion of worts into beers and ales.

Product.	Loss in solids.	Loss in sugar.	Alcohol.	Loss in solids divided by alcohol.	Difference between loss in solids and loss in sugar.
	Grams per 100 cc.	Grams per 100 cc.	Grams per 100 cc.		Grams per 100 cc.
Beer (all-malt)	8.59	8.45	3.85	2.23	0.14
Do	8.55	8.43	3.91	2.18	.12
Do	8.27	8.49	3.83	2.13	.22
Do	8.48	8.27	3.72	2.27	.21
Do	8.02	7.99	3.66	2.19	.03
Beer (60 per cent malt and 40 per cent corn)	7.27	7.42	3.45	2.10	.15
Do	7.53	7.73	3.33	2.00	.20
Beer (80 per cent malt and 20 per cent rice)	7.35	7.46	3.31	2.22	.11
Do	7.07	7.13	3.18	2.22	.06
Beer (all-malt)	6.04	6.05	3.02	2.00	.01
Do	5.86	6.01	2.87	2.04	.15
Average for beers				2.14	.04
Porter (small)	9.23	7.60	4.30	2.14	1.63
Porter (large)	9.91	8.32	4.70	2.10	1.59
Do	10.06	8.27	4.75	2.11	1.79
Do	9.90	7.82	4.72	2.09	2.08
Ale	10.31	8.84	4.82	2.13	1.47
Do	10.63	8.98	5.11	2.08	1.65
Do	10.70	9.20	4.93	2.17	1.50
Do	11.15	8.94	5.16	2.15	2.21
Do	11.50	9.57	5.34	2.15	1.93
Do	11.62	9.06	5.35	2.17	2.56
Pale ale	11.04	9.50	5.43	2.03	1.54
Do	11.88	9.63	5.67	2.09	2.25
Do	11.75	9.11	5.53	2.12	2.64
Brown stout	12.03	9.38	5.69	2.11	2.65
Average for ales				2.12	1.96
Average for beers and ales				2.13	

TABLE VI.—Average loss during fermentation.

Kind of wort.	Protein.	Ash.	Phosphoric acid.
	<i>Grams per 100 cc.</i>	<i>Grams per 100 cc.</i>	<i>Grams per 100 cc.</i>
Beer worts (all-malt).....	0.209	0.017	0.015
Beer worts (malt and rice).....	.210	.015	.014
Beer worts (malt and corn).....	.168	.014	.013
Porter worts.....	.213	.031	.017
Ale worts.....	.275	.029	.019

[back](#)

TABLE VII.—Analyses of beers and ales from various breweries.

Sample No.	Raw materials.	Product.	Alcohol.	Extract (Schultz and Ostermann).	Extract in original wort (calculated).	Degree of fermentation.	Total acid as lactic.	Volatile acid as acetic.	Reducing sugars as anhydrous maltose.	Dextrin.	Protein (N × 6.25).	Ash.	Phosphoric acid (as P ₂ O ₅).	Undetermined.	Polarimeter.	Color (Lovibond) in 1-inch cell.	Calculated to basis of wort with 15 per cent solids.				
																	Protein (N × 6.25).	Ash.	Phosphoric acid (as P ₂ O ₅).		
Brewery No. 1.																					
14004-H	Malt.....	Beer.....	P. ct. by wght. 3.07	P. ct. 5.55	P. ct. 11.69	32.54	P. ct. 0.223	P. ct. 0.013	P. ct. 1.59	P. ct. 2.64	P. ct. 0.615	P. ct. 0.207	P. ct. 0.072	P. ct. 0.49	Degs. V. +58.0	Degs. ¹ 5	P. ct. 0.789	P. ct. 0.266	P. ct. 0.092		
14005-H	65 per cent malt and 35 per cent cereal.	...do....	2.42	6.19	11.03	43.88	.142	.013	1.81	3.15	.355	.141	.042	.73	+50.8	3	.483	.192	.037		
14006-H	60 per cent malt and 40 per cent corn.	...do....	2.16	5.28	9.60	45.00	.178	.007	1.52	2.96	.260	.124	.039	.42	+44.0	4	.407	.194	.061		
Brewery No. 2.																					
22017-D	Malt.....	Beer.....	3.83	5.06	12.72	60.22	.214	.001	1.32	2.30	.603	.206	.079	.63		12	.712	.243	.093		
22018-D	do.....	do.....	3.90	5.06	12.86	60.65	.223	.002	1.34	2.03	.606	.199	.077	.88		13	.701	.230	.090		
22020-D	do.....	do.....	3.69	5.12	12.50	59.01	.231	.002	1.57	2.20	.630	.203	.080	.52		11	.756	.244	.096		
22021-D	do.....	do.....	3.63	5.54	12.80	56.72	.219	.001	1.82	2.15	.626	.205	.081	.74		10	.734	.240	.095		
22012-D	80 per cent malt and 20 per cent rice.	...do....	3.16	5.13	11.45	55.20	.241	.003	1.55	2.48	.395	.154	.056	.55	+37.2	2	.517	.202	.073		
Brewery No. 3.																					
29517-B	80 per cent malt and 20 per cent cereal.	Ale.....	6.33	6.77	19.43	65.15	.357	.005	1.42	3.26	.622	.279	.066	.75	+40.4		.480	.215	.051		
22027-D	78 per cent malt and 22 per cent cereal.	...do....	5.14	5.82	16.10	63.35	.205	.004	1.23	3.27	.488	.189	.054	.64	+41.2	3	.455	.176	.050		
22040-D	do.....	do.....	5.31	6.17	16.79	63.25	.291	.001	1.40	3.29	.533	.189	.050	.76	+38.6	4	.476	.169	.045		
22045-D	do.....	do.....	5.33	6.15	16.81	63.42	.232	.002	1.51	3.29	.563	.203	.045	.58	+40.4	4	.502	.181	.040		
29512-B	75 per cent malt and 25 per cent cereal.	...do....	5.32	8.88	19.52	54.51	.366	.008	2.62	4.28	.650	.266	.057	1.06	+65.2	3	.499	.204	.044		
29514-B	do.....	do.....	5.43	8.25	19.11	56.83	.357	.005	2.45	3.99	.650	.250	.056	.91	+59.8	2	.500	.196	.044		
29519-B	do.....	do.....	5.43	8.70	19.56	55.52	.278	.006	2.56	4.21	.655	.249	.056	1.02	+64.0	2	.502	.191	.043		
22030-D	65 per cent malt, 28 per cent cereal, and 7 per cent brewer's sugar.	...do....	4.79	5.80	15.38	62.29	.223	.003	1.14	3.20	.419	.190	.038	.85	+40.0	5	.409	.185	.037		
22039-D	do.....	do.....	5.10	5.55	15.75	64.64	.223	.003	1.24	2.84	.465	.191	.042	.37	+37.6	5	.443	.175	.040		
22043-D	do.....	do.....	4.91	5.50	15.32	63.27	.214	.003	1.38	2.81	.436	.170	.042	.70	+38.0	4	.427	.166	.011		

¹ Brewer's scale.[back](#)

TABLE VIII.—Summary of the results of analyses (showing ash, protein, and phosphoric acid determinations) in all finished products of known composition, calculated to the basis of a uniform wort containing 15 per cent of solids.

Raw materials.	Products.	Ash.	Protein (N×6.25).	Phos- phoric acid (as P ₂ O ₅).
Malt.....	Beers: 21 samples:	<i>Per cent.</i>	<i>Per cent.</i>	<i>Per cent.</i>
	Maximum.....	0.336	1.079	0.143
	Minimum.....	.230	.701	.087
	Average.....	.275	.870	.109
80 per cent malt and 20 per cent rice.....	Beer.....	.202	.517	.073
66 per cent malt and 34 per cent rice.....	do.....	.198	.555	.084
62 per cent malt and 38 per cent rice.....	do.....	.205	.488	.061
55 per cent malt and 45 per cent rice.....	do.....	.148	.380	.077
50 per cent malt and 50 per cent rice.....	do.....	.167	.351	.056
	Maximum...	.205	.555	.084
70 per cent malt and 30 per cent corn.....	Beer.....	.199	.343	.057
Do.....	do.....	.188	.367	.065
68 per cent malt and 32 per cent corn.....	do.....	.150	.461	.057
Do.....	do.....	.181	.466	.062
Do.....	do.....	.164	.459	.056
60 per cent malt and 40 per cent corn.....	do.....	.215	.563	.074
Do.....	do.....	.188	.593	.076
Do.....	do.....	.223	.597	.074
45 per cent malt and 55 per cent corn.....	do.....	.145	.347	.057
	Maximum...	.223	.597	.076
65 per cent malt and 35 per cent cereal in.....	Beer.....	.192	.483	.057
80 per cent malt and 20 per cent cereal in.....	Ale.....	.215	.480	.051
78 per cent malt and 22 per cent cereal in.....	do.....	.176	.455	.050
Do.....	do.....	.169	.476	.045
Do.....	do.....	.181	.502	.040
75 per cent malt and 25 per cent cereal in.....	do.....	.204	.499	.044
Do.....	do.....	.196	.509	.044
Do.....	do.....	.191	.502	.043
65 per cent malt, 7 per cent brewer's sugar, and 28 per cent cereal in.....	do.....	.185	.409	.037
Do.....	do.....	.175	.443	.040
Do.....	do.....	.166	.427	.041
	Maximum...	.213	.509	.051

[back](#)

TABLE IX.—Analyses of all-malt American beers.

Sample No.	Alcohol.	Extract (Schultz and Ostermann).	Extract in original wort (calculated).	Degree of fermentation.	Total acid as lactic.	Volatile acid as acetic.	Reducing sugars as anhydrous maltose.	Dextrin.	Protein (N×6.25).	Ash.	Phosphoric acid (as P ₂ O ₅).	Undetermined.	Polarimeter.	Color (Lovibond) in ½ inch cell.	Calculated to basis of wort with 15 per cent of solids.		
															Protein (N×6.25).	Ash.	Phosphoric acid (as P ₂ O ₅).
	Per cent by weight.	Per cent.	Per cent.		Per cent.	Per cent.	Per cent.	Per cent.	Per cent.	Per cent.	Per cent.	Per cent.	Degrees V.	Degrees.	Per cent.	Per cent.	Per cent.
22017-D...	3.83	5.06	12.72	60.22	0.214	0.001	1.32	2.30	0.603	0.206	0.079	0.63		12	0.712	0.243	0.093
22018-D...	3.90	5.06	12.86	60.65	.223	.002	1.34	2.03	.606	.199	.077	.88		13	.701	.230	.090
22020-D...	3.69	5.12	12.50	59.04	.234	.002	1.57	2.20	.630	.203	.080	.52		11	.756	.244	.096
22021-D...	3.63	5.54	12.80	56.72	.219	.001	1.82	2.15	.626	.205	.081	.74		10	.734	.240	.095
16289-C...	3.00	5.88	11.88	50.50	.228	.003	1.33	2.68	.752	.242	.081	.88	+36.8		.950	.306	.102
16299-C...	2.84	6.02	11.70	48.55	.232	.012	1.38	2.77	.724	.237	.088	.91	+40.0		.932	.304	.113
20714-D...	3.07	5.80	11.94	51.42	.241	.012	1.45	2.64	.721	.225	.089	.76	+36.4		.906	.283	.112
20715-D...	2.95	5.77	11.67	50.56	.228	.009	1.43	2.67	.725	.213	.088	.73	+36.0		.932	.274	.113
23571-E...	3.68	4.44	11.80	62.45	.232	.010	1.06	1.67	.653	.229	.096	.83	+21.6		.830	.291	.122
23585-E...	3.60	5.04	12.24	58.82	.277	.005	1.36	1.81	.811	.257	.102	.80	+23.2		.994	.315	.124
23528-E...	3.28	6.36	12.92	50.80	.384	.016	1.62	2.74	.905	.239	.123	.86	+37.2		1.051	.277	.143
23533-E...	3.41	5.48	12.30	55.45	.232	.008	1.48	2.44	.612	.200	.086	.75	+33.6		.747	.244	.105
23537-E...	3.80	7.26	14.86	51.35	.250	.012	2.51	2.95	.802	.228	.098	.77	+44.0		.809	.230	.099
23538-E...	3.16	6.11	12.43	50.84	.250	.008	1.93	2.41	.797	.208	.086	.76	+37.2		.952	.251	.104
23538-E...	3.13	6.61	12.77	48.24	.250	.009	2.13	2.82	.612	.225	.087	.82	+43.6		.719	.264	.103
23539-E...	3.35	6.21	12.91	51.90	.178	.017	1.78	2.87	.627	.204	.075	.73	+40.0		.729	.237	.087
23539-E...	3.22	6.63	13.27	48.53	.312	.017	2.18	2.58	.778	.248	.097	.84	+38.8		.879	.280	.109
23540-E...	3.93	6.77	14.63	53.73	.348	.007	2.64	1.87	1.010	.324	.129	.93	+34.0		1.035	.332	.132
23590-E...	3.48	5.45	12.41	56.09	.375	.010	2.21	1.48	.892	.264	.109	.60	+25.6		1.079	.319	.132
23541-E...	3.12	5.06	11.30	55.22	.259	.004	1.69	1.27	.777	.253	.087	1.07	+20.0		1.031	.336	.115
14004-H...	3.07	5.55	11.69	52.54	.223	.013	1.59	2.64	.615	.207	.072	.49		5	.789	.266	.092

¹ Brewer's scale.

[back](#)

TABLE X.—Analyses of malt-and-rice American beers.

Sample No.	Raw materials.	Alcohol.	Extract (Schultz and Ostermann).	Extract in original wort (calculated).	Degree of fermentation.	Total acid as lactic.	Volatile acid as acetic.	Reducing sugars as anhydrous maltose.	Dextrin.
		<i>Per ct. by weight.</i>	<i>Per ct.</i>	<i>Per ct.</i>		<i>Per ct.</i>	<i>Per ct.</i>	<i>Per ct.</i>	<i>Per ct.</i>
22042-D	80 per cent malt and 20 per cent rice.....	3.16	5.13	11.45	55.20	0.241	0.003	1.55	2.41
23527-E	66 per cent malt and 34 per cent rice.....	3.32	5.50	12.14	54.70	.196	.008	1.46	2.74
23581-E	62 per cent malt and 38 per cent rice.....	2.86	6.16	11.88	48.16	.178	.014	1.77	3.15
23587-E	55 per cent malt and 45 per cent rice.....	3.56	4.96	12.08	53.94	.151	.007	1.24	2.53
23586-E	50 per cent malt and 50 per cent rice.....	3.44	5.67	12.55	54.82	.160	.008	1.44	3.08

Sample No.	Raw materials.	Protein (N×6.25).	Ash.	Phosphoric acid (as P ₂ O ₅).	Undetermined.	Polarimeter.	Color (Lovibond) in 1-inch cell.	Calculated to basis of wort with 15 per cent of solids.		
								Protein (N×6.25).	Ash.	Phosphoric acid (as P ₂ O ₅).
		<i>Per ct.</i>	<i>Per ct.</i>	<i>Per ct.</i>	<i>Per ct.</i>	<i>Degrees V.</i>	<i>Degrees.¹</i>	<i>Per ct.</i>	<i>Per ct.</i>	<i>Per ct.</i>
22042-D	80 per cent malt and 20 per cent rice.....	0.395	0.154	0.056	0.62	+37.2	2	0.517	0.202	0.073
23527-E	66 per cent malt and 34 per cent rice.....	.449	.160	.068	.69	+40.0	2	.555	.198	.034
23581-E	62 per cent malt and 38 per cent rice.....	.386	.162	.048	.69	+48.6		.488	.205	.061
23587-E	55 per cent malt and 45 per cent rice.....	.306	.119	.062	.76	+37.0	2	.380	.148	.077
23586-E	50 per cent malt and 50 per cent rice.....	.294	.140	.047	.72	+46.4	2	.351	.167	.056

¹ Brewer's scale.

[back](#)

TABLE XI.—Analyses of malt-and-corn American beers.

Sample No.	Raw materials.	Alcohol.	Extract (Schultz and Ostermann).	Extract in original wort (calculated).	Degree of fermentation.	Total acid as lactic.	Volatile acid as acetic.	Reducing sugars as anhydrous maltose.	Dextrin.
		<i>Per ct. by weight.</i>	<i>Per ct.</i>	<i>Per ct.</i>		<i>Per ct.</i>	<i>Per ct.</i>	<i>Per ct.</i>	<i>Per ct.</i>
3534-E	70 per cent malt and 30 per cent corn.....	2.75	5.53	11.03	50.32	0.125	0.005	1.27	3.13
23535-E	do.....	3.03	4.85	10.91	55.55	.224	.011	1.15	2.76
23518-E	68 per cent malt and 32 per cent corn.....	3.37	5.96	12.70	53.07	.116	.006	1.03	3.07
23561-E	do.....	3.26	6.14	12.66	51.50	.134	.016	1.54	3.07
23572-E	do.....	3.37	6.07	12.81	52.62	.143	.020	1.56	3.29
23584-E	60 per cent malt and 40 per cent corn.....	3.09	4.90	11.08	55.78	.178	.020	1.35	2.43
23523-E	do.....	3.16	6.07	12.39	51.01	.214	.009	1.60	3.36
23660-E	do.....	3.26	6.07	12.59	51.79	.205	.009	1.55	3.06
16236-C	do.....	3.19	5.60	11.98	53.34	.173	.014	1.85	2.61
16247-C	do.....	3.23	5.67	12.13	53.26	.178	.014	1.84	2.41
23524-E	45 per cent malt and 55 per cent corn.....	3.43	5.75	12.61	54.40	.169	.007	1.44	3.19

Sample No.	Raw materials.	Protein (N×6.25).	Ash.	Phosphoric acid (as P ₂ O ₅).	Undetermined.	Polarimeter.	Color (Lovibond) in 1-inch cell.	Calculated to basis of wort with 15 per cent of solids.		
								Protein (N×6.25).	Ash.	Phosphoric acid (as P ₂ O ₅).
		<i>Per ct.</i>	<i>Per ct.</i>	<i>Per ct.</i>	<i>Per ct.</i>	<i>De-grees V.</i>	<i>De-grees.¹</i>	<i>Per ct.</i>	<i>Per ct.</i>	<i>Per ct.</i>
23534-E	70 per cent malt and 30 per cent corn.....	0.252	0.146	0.042	0.73	+42.6	2	0.343	0.199	0.057
23535-E	do.....	.267	.137	.047	.53	+37.2	2	.367	.188	.065
23518-E	68 per cent malt and 32 per cent corn.....	.390	.127	.048	.74	+49.6	3	.461	.150	.057
23561-E	do.....	.393	.153	.052	.98	+48.0		.466	.181	.062
23572-E	do.....	.392	.140	.048	.69	+47.0		.459	.164	.056
23584-E	60 per cent malt and 40 per cent corn.....	.416	.159	.055	.54	+34.6		.563	.215	.074
23523-E	do.....	.490	.155	.064	.46	+45.0	5	.593	.188	.076
23660-E	do.....	.501	.187	.062	.77	+44.4	5	.597	.223	.074
16236-C	do.....	.311	.219	.057	.61	+41.0		.389	.274	.071
16287-C	do.....	.322	.205	.055	.89	+41.0		.398	.254	.068
23524-E	45 per cent malt and 55 per cent corn.....	.292	.122	.048	.71	+45.6	2	.347	.145	.057

¹ Brewer's scale.[back](#)

TABLE XII.—Analyses of American beers incorrectly represented to be all-malt.

Sample No.	Alcohol.	Extract (Schultz and Ostermann).	Extract in original wort (calculated).	Degree of fermentation.	Total acid as lactic.	Volatile acid as acetic.	Reducing sugars as anhydrous maltose.	Dextrin.	Protein (N×6.25).	Ash.	Phosphoric acid (as P ₂ O ₅).	Undetermined.	Polarimeter.	Color (Lovibond) in 1-inch cell.	Calculated to basis of wort with 15 per cent of solids.		
															Protein (N×6.25).	Ash.	Phosphoric acid (as P ₂ O ₅).
	Per cent by weight.	Per cent.	Per cent.		Per ct.	Per cent.	Per cent.	Per ct.	Per cent.	Per ct.	Per cent.	Per cent.	Degrees V.	Degrees. ¹	Per cent.	Per ct.	Per cent.
2417-E	3.67	4.66	12.00	61.33	0.154	0.013	1.59	1.97	0.341	0.120	0.053	0.64	+31.0	1	0.426	0.150	0.066
36612-E	3.10	7.55	13.75	45.09	.232	.017	2.33	3.93	.274	.149	.043	.87	+54.4	35	.299	.163	.047
1146-E	4.23	7.35	15.81	53.51	.241	.029	2.21	3.39	.472	.202	.074	1.08	+50.0	32	.448	.192	.070
3014-E	3.26	6.74	13.29	49.17	.167	.014	2.74	2.56	.406	.147	.060	.89	+49.6	3	.459	.166	.068
2538-E	2.86	6.69	12.41	46.09	.214	.017	3.22	2.12	.390	.181	.050	.78	+43.2	54	.472	.219	.060
1734-E	3.55	6.55	13.65	52.02	.250	.028	1.87	3.58	.303	.145	.048	.65	+53.0	3	.333	.159	.053
1154-E	3.55	5.62	12.72	55.82	.187	.014	1.88	2.52	.352	.158	.043	.71	+41.0	3	.415	.186	.051
1006-E	3.58	4.74	11.90	60.17	.138	.017	1.42	1.99	.420	.158	.057	.75	+31.2	12	.529	.199	.072
195-E	3.56	4.21	11.33	62.84	.127	.008	2.72	.77	.254	.127	.044	.34	+22.6	3	.336	.168	.058
5022-E	2.95	5.60	11.50	51.30	.133	.022	2.30	2.20	.233	.121	.036	.75	+41.0	3	.304	.158	.047
1005-E	2.69	5.33	10.71	50.24	.102	.010	1.77	2.34	.354	.132	.048	.74	+39.6	3	.496	.185	.067
194-E	3.18	4.00	10.36	61.39	.062	.017	1.37	1.58	.296	.110	.045	.64	+27.0	5	.423	.159	.065
185-E	3.44	3.92	10.80	63.70	.141	.011	2.06	.97	.250	.116	.041	.52	+23.2	3	.347	.161	.057
5314-E	3.78	5.65	13.21	57.15	.151	.010	1.46	2.84	.416	.177	.059	.76	+40.8	3	.472	.201	.067
4451-E	3.23	6.52	12.98	49.77	.228	.009	2.80	2.44	.519	.153	.054	.61		78	.599	.177	.062
5096-E	4.19	6.44	14.82	56.55	.268	.012	2.15	3.06	.309	.197	.082	.72	+44.0	18	.313	.199	.083
3359-E	4.96	7.43	17.35	57.18	.321	.019	1.76	3.74	.509	.294	.074	1.13	+48.4	44	.440	.254	.084
3358-E	3.23	5.53	11.99	53.83	.121	.009	1.88	2.51	.314	.175	.045	.65	+41.0	3	.392	.219	.056
481-E	3.87	5.20	12.94	59.82	.147	.014	1.32	2.47	.319	.163	.054	.93	+36.6	2	.370	.189	.063
687-E	3.24	4.71	11.19	57.91	.080	.009	1.63	2.20	.284	.124	.048	.47	+35.6	2	.382	.167	.065
193-E	3.66	5.75	13.07	56.01	.148	.017	1.78	2.67	.346	.153	.048	.80	+42.8	3	.397	.176	.055
5023-E	3.16	5.30	11.62	54.39	.128	.005	1.95	2.13	.214	.126	.033	.88	+40.0	2	.276	.163	.043
5318-E	3.57	5.20	12.34	57.86	.220	.003	1.50	2.59	.320	.147	.045	.64	+41.8	3	.389	.179	.055
6715-E	3.75	7.98	15.48	48.45	.174	.006	2.55	3.60	.430	.186	.058	1.21	+61.6	14	.417	.180	.056
6716-E	3.66	7.73	15.05	48.64	.187	.009	2.37	3.84	.384	.164	.055	.97	+61.6	14	.383	.164	.055
2388-E	2.95	5.35	11.25	52.45	.080	.012	1.64	2.32	.337	.135	.042	.91	+40.4	2	.449	.180	.056
2352-E	3.20	4.69	11.09	57.71	.147	.011	1.99	1.76	.328	.146	.037	.46	+32.2	3	.441	.198	.050
2770-E	3.66	5.99	13.31	55.00	.169	.019	1.82	2.56	.278	.194	.050	1.14	+45.2	6	.311	.219	.056
8705-E	2.91	5.62	11.44	50.96	.147	.017	1.86	2.58	.366	.150	.041	.66	+41.0		.480	.197	.053
8706-E	3.48	4.93	11.89	58.54	.214	.022	1.83	1.83	.470	.160	.062	.64	+30.4		.593	.202	.078
8704-E	2.76	4.98	10.50	52.57	.160	.010	1.62	2.26	.390	.150	.054	.56	+36.4		.557	.214	.077
7839-E	3.51	5.38	12.40	56.61	.142	.012	1.57	2.77	.290	.116	.031	.63	+41.2		.351	.140	.038
5707-E	3.35	5.82	12.52	53.51	.160	.011	2.01	2.55	.352	.147	.048	.66	+43.0		.422	.176	.058
8171-E	3.50	5.99	12.99	53.85	.169	.013	1.92	2.67	.271	.178	.054	.95	+45.2		.313	.205	.062
397-E	3.73	6.71	14.17	52.65	.169	.007	1.56	3.96	.324	.151	.051	.72	+55.2	2	.343	.160	.054

¹ Brewer's scale.[back](#)